

# Boundary Theorems and Bench Findings: Reconciling the CAROM Experimental Program with the Oruži Formal Corpus

Working note — CAROM series

July 2026

## Abstract

Four Lean-formalized working notes (architecture cost boundaries; predictive coding exactness and cost; OEIS verified synthesis with a four-carrier campaign; GSLT typed-action decoding) prove boundary theorems for mechanisms the CAROM experimental program has been measuring empirically. This note reconciles the two bodies of work for two audiences. For the formal side: a ledger of which bench findings your theorems explain, which they reframe, and which of our empirical laws are candidates for your formalization pipeline — including one place where our data suggests a hypothesis your affine theory makes precise (learned commutation as a third reading of the itinerary-tau decline). For the implementation side: a probe catalog — delivered as running code (`harness.v3_probes.py`) — with, for each probe, the theorem that licenses it, what it measures, how to call it, and how to read the numbers; plus the concrete rescoping of PC-CAROM, the reachability discipline for the teach loop, and the portfolio reframing of the execution-semantics ladder. Smoke readings on the sandbox checkpoint are included throughout, one of which is already substantive: the learned phase family fails schedule-uniform stability under adversarial alternation (norm growth  $3.8\text{--}4.9\times$  over 16 phases; joint-spectral-radius proxy  $p_{90} = 1.07$ ) while natural trajectories behave — “stability does not compose,” observed on our own artifact.

## 1 Purpose and audiences

**Audience A** (Zar and the Oruži collaborators): §2 is the correspondence ledger; §3 the new experiment your commutation theory licenses; §8 the conjecture list flowing back to Lean. **Audience B** (the coding agents implementing the CAROM experiments): §4 is the operational probe catalog; §§5–7 the program changes, each with the theory result that motivates it stated in implementation terms. Read §3 either way — it is the highest-information next run and it uses only existing checkpoints.

Throughout, `name` refers to a Lean declaration in the Oruži corpus; function names in `monospace` refer to `harness.v2.py` and `harness.v3_probes.py`.

## 2 The correspondence ledger

| Bench finding (ours)                                                                                                                | Theorem (Oruži)                                                                                                                                                                     | Status                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Routing bottleneck: realizable instruction set $\approx K$ ; 210 slot-addressed commands stall through $K=4$ operators              | <code>mixedExpertStep_eq_singleInterpolation</code><br>simplex mixture of gated-interpolation experts is one gated interpolation — routing buys selectivity, not a new update class | Empirical law now has its mechanism at update-class level                                            |
| Opaque tokens fail paraphrase transfer; LM-anchored routing transfers where representation similarity exists                        | <code>commandTrajectory_eq_of_mixtureSchedule</code><br>commands carry no semantics; meaning lives in routes, not names                                                             | Dead-end doctrine confirmed: semantics must enter through the route map                              |
| Routing interference: paired $\Delta = -0.144$ after registration, 94% of degraded cases carry routing leak, all old weights frozen | Consolidation/action-space results; “frozen weights $\neq$ frozen behavior” when a shared softmax grows                                                                             | Independently arrived at both sides; our paired-M2 number is the quantitative instance               |
| KL no-regression penalty at skill registration (specified, untested)                                                                | Faithfulness-is-reachability<br>( <code>faithfulConsolidation_iff_effective_rank_bottleneck</code> corollary<br>( <code>rankBudgetReachable_iff_rank_le</code> ))                   | Penalty is a minimum-change update; check reachability <i>before</i> tuning its weight (§6)          |
| DEQ contraction penalty; convergence-depth law (routing gradient collapses first at insufficient settling)                          | $(P-1)^2$ law; $\rho(I-P) < 1$ membership condition; depth cost $\cos(\pi/n)$ ; first-arrival attenuation                                                                           | Same phenomena; their exact rates calibrate our residual-criterion settling                          |
| EqProp = BPTT parity at matched steps and wall-clock (toy, energy-based core)                                                       | PC verdict: dead as drop-in; survives on energy-based substrates, frozen caps, staged ignition                                                                                      | Our result sits inside their surviving region; scoping in §5                                         |
| Tau declines $0.990 \rightarrow 0.492$ while accuracy rises                                                                         | <code>linearPhases_orderIndependent_iff_affinePhases_orderIndependent_if</code><br>a third hypothesis and the discriminating measurement (§3)                                       | Interference energy zero; if                                                                         |
| GLV channel tuned and stable on natural schedules                                                                                   | <code>individuallyStable_switchingDivergence</code><br><code>commutingCommonQuadraticLyapunov</code>                                                                                | Res-phase certificates wordiness; smoke probe already exhibits non-schedule-uniform stability (§4.4) |
| Single-seed instability (sq paraphrase 0.44 vs 0.28); audit’s fixed-corpus mandate                                                  | Telemetry non-identifiability: nine of eleven diagnostics confounded; attribution requires enumerated intervention                                                                  | harness.v2 philosophy stated as theorems; their probe list completes our v3 (§4)                     |
| Exp2 evaluation logged loss/accuracy only                                                                                           | <code>equal_crossEntropy_strictly_order</code><br>fit does not imply guidance; decisiveness is the operative quantity                                                               | Decisiveness Yields now standard in every arm ( <code>decisiveness_report</code> )                   |

### 3 The tau three-hypothesis experiment

The audited GPU run’s central anomaly — itinerary tau falling from 0.990 to 0.492 while endpoint accuracy rose — had two candidate readings: *mechanistic drift* (the model abandons itinerant execution for distributed mixing) and *metric artifact* (early tau rewarded ordered subsets; demonstrated live by harness\_v2’s self-test and by the smoke checkpoint: tau 0.997 with coverage 0.74, exact-order 0.36). The commutation theory supplies a third:

**Learned commutation.** Training drives the effective per-phase maps toward pairwise commutation (and vanishing affine obstruction), so execution order *stops mattering*. The channel can then smear or reorder at zero accuracy cost: tau falls not because the mechanism decays but because its necessity dissolves.

The three hypotheses are separated by measuring, per checkpoint, four quantities: repaired itinerary metrics (coverage, exact-order, transition recall), the intervention `causal_gap` (forced-correct minus shuffled endpoint accuracy), the linearized pairwise interference energy  $\mathbb{E} \| [J_a, J_b] v \|^2$  (`commutator_energy`, via JVP probes — an intervention, as the identifiability results require), and the exact nonlinear `swap_discrepancy`  $\| \Phi_b \Phi_a w - \Phi_a \Phi_b w \| / \| w \|$ , which is the ground truth the affine formula  $[B, A]x + (Ba + b - Ab - a)$  linearizes. Decision table:

|                     | coverage/exact | causal_gap | interference/swap        |
|---------------------|----------------|------------|--------------------------|
| metric artifact     | stays high     | stays high | (any)                    |
| mechanistic drift   | falls          | falls      | <b>stays high</b>        |
| learned commutation | falls          | falls      | <b>falls in lockstep</b> |

Drift with high interference is the alarming case (order matters and the model gets answers another way); learned commutation is the benign and theoretically interesting case — and if observed, the same Gram certifies *which* command pairs a compiled schedule may treat as a set, directly reducing what the Exp2 edge compiler must get right. The sweep is implemented as `tau_three_hypothesis_sweep(ckpt_paths, ...)` and runs read-only on the audited run’s saved checkpoints.

### 4 Probe catalog (audience B)

All probes live in `harness_v3_probes.py`, are read-only over checkpoints, and were smoke-verified on the sandbox artifact (checkpoint at  $\sim 213$  updates — early regime; treat the numbers as demonstrations of the instruments, not results). Raise `n/probes` on GPU.

#### 4.1 `swap_discrepancy` / `commutator_energy`

**License:** `affinePhases_orderIndependent_iff, ...interferenceEnergy_zero`. **Measures:** exact and linearized order-sensitivity of command-phase pairs. **Implementation notes:** `phase_map` is the differentiable single-phase executor (control clamped, amplitude 1.5, dwell 12 steps); Jacobian–vector products via `torch.func.jvp` — never materialize the Jacobian. **Smoke:** swap median 0.106; interference mean  $5.2 \times 10^{-3}$ . **Read:** track *trends across checkpoints*, not absolute values; the pairwise matrix (extendable from the mean) is the Gram that certifies reorderable schedule pairs.

## 4.2 tau\_three\_hypothesis\_sweep

As §3. Output: one row per checkpoint with accuracy, repaired itinerary metrics, causal\_gap, interference, swap discrepancy. Plot all columns against training step; apply the decision table.

## 4.3 schedule\_stability\_probe

**License:** `individuallyStableSwitchingDiverges`. **Measures:** (a) workspace norm growth under adversarial two-command alternation; (b) a joint-spectral-radius proxy from random products of phase JVPs. **Smoke (substantive):** alternation growth median  $3.85\times$  (max  $4.85\times$ ) over 16 phases; JSR proxy median 1.0001,  $p_{90}$  1.074  $\Rightarrow$  *not schedule-uniform stable*, though natural trajectories behave. **Read:** natural-schedule health certifies nothing about command programs; if the compiled channel will ever execute model-chosen or adversarial schedules, either constrain the phase family toward a common Lyapunov certificate (`commutingCommonQuadraticLyapunovCrown`: commuting stable generators always admit one — commutation again) or bound schedules to the certified set.

## 4.4 cross\_slot\_coupling\_probe

**License:** `crossSlotHessianNotIdentifiable` — the one quantity that would justify typed slots is provably invisible in logs; only perturbation reveals it. **Measures:** normalized coupling matrix  $C_{ij}$  (readout effect on slot  $i$  of perturbing slot  $j$ ’s evidence). **Smoke:** off-diagonal mass 0.60 — strongly coupled, correctly reflecting that `rev/shr/cms` are cross-slot operations; per-slot independent treatment is *not* licensed on this task family. **Read:** any claimed typed-slot advantage must ship with this matrix, or the attribution is unsupported.

## 4.5 decisiveness\_report

**License:** `equalCrossEntropyStrictlyOrderedExpectedYield`. **Measures:** routing entropy per command (vs.  $\log K$ ) and normalized control-state entropy over time. **Smoke:** routing entropy 1.29 against uniform 2.08. **Read:** report beside every loss/accuracy; loss provably underdetermines guidance-like properties, and our own tau-vs-accuracy divergence was an instance.

# 5 Rescoping PC-CAROM (audience B, with A checking the scoping)

Our EqProp-vs-BPTT parity result stands, *scoped*: energy-based substrate (the EqProp-native regime), toy scale (below the measured  $1.6\text{--}15\times$  settling cost), matched steps. The Oruži PC verdict — dead as a drop-in full-network rule on synchronous hardware; alive for shallow adapters over *exactly* frozen backbones, staged BP $\rightarrow$ PC ignition, rank-uniform skips, and asynchronous hardware — dictates three changes: (1) PC-CAROM training runs start from BPTT-trained checkpoints (staged ignition), which our checkpointing supports for free; (2) the target regime is the *cap*: thin PC-trained CAROM over a frozen LLM, which is precisely the frozen-backbone-restriction regime their theorem endorses — with the hard warning that partial freezing or shared active normalization re-exports the backbone depth bound, so the freeze must be exact; (3) our convergence-depth law (routing alignment collapses first at insufficient settling) is now calibrated

by their exact rates ( $\cos(\pi/n)$  per sweep;  $(P-1)^2$  per step;  $0 < \eta\lambda < 2$  coupled window): implement residual-criterion settling, and expect required depth to grow as training stiffens the landscape.

## 6 The teach loop under the consolidation theory (audience B)

Map: registering a taught skill = growing a shared action space over frozen components. Their results give the discipline: (i) *reachability first* — before tuning the KL no-regression weight, check whether “preserve old-command routing + install new-skill routing” is jointly reachable in the gate parameters (`exists_faithfulConsolidation_iff_mem_range`; rank version `rankBudgetReachable_iff_rank_le`): if not, no penalty weight fixes it and the gate needs capacity, not tuning. (ii) The KL penalty is a minimum-change update in the GEM/OGD family their reachability criterion licenses — implement it as distillation against *cached pre-registration routing distributions* on a frozen old-command corpus, not as new-slot suppression. (iii) Rehearsal from the generative teacher is free, but rehearsed evidence must be *discounted*: re-counting a reused contribution overstates by exactly one copy (`reused_contribution_exact_excess`). (iv) Our paired-M2 decomposition (routing-leak vs. operator attribution) is the measurement their “frozen weights  $\neq$  frozen behavior” statement calls for; keep it mandatory at every registration.

## 7 Portfolio reframing and the belief-register arm

The OEIS four-carrier campaign (984 discoveries; 669 unique to one carrier; pairwise overlap 8–16%; fixed split 399 vs. best arm 435; adaptive allocation dominant; the last-place belief arms producing two of five showcase-proved programs) converts our scheduled/fixed-point/itinerant ladder from a contest into a portfolio. Implementation changes: report union value and per-arm unique coverage, not only per-arm accuracy; allocate adaptively; and add a fifth arm — **belief-register CAROM**: workspace slots as natural-coordinate evidence registers  $(\eta, \Lambda)$  or counts, the write hardwired as addition (their fusion-is-addition square: Kalman = gated interpolation = PLN revision = addition in natural coordinates), learning concentrated where Bayes is silent — measurement maps and gain schedules — and, under drift, fading at the *derived* retention  $P/(P+Q)$  rather than a tuned constant. Expectation set by their data: likely lower raw yield, likely complementary coverage, and calibrated per-slot confidence for free (the principled version of our residual-as-confidence idea).

## 8 What flows back to the formal side (audience A)

Your affine/linear phase theory stops exactly where our program’s distinctive content begins. Candidates for the Lean pipeline, with fixtures available from our artifacts:

- (C1) **Heteroclinic execution.** GLV channels, saddle-value conditions  $\nu > 1$ , dwell-time law  $\ln(1/\eta)/\lambda$ , fatigue as anti-trap, entry-by-initialization — none covered by the affine results. Our tuning package is a conjecture set with measured constants.
- (C2) **The routing-bottleneck bound, sharpened.** Your collapse theorem gives the update class; the quantitative version — realizable distinct behaviors under sharp routing =  $K$  vertices, with a defect bound for soft mixtures — would turn our design rule into a theorem with a constant.

- (C3) **Routing interference under action-space growth.** A softmax-renormalization analog of your consolidation double-counting result: exact leak formula for old-command probability mass as a function of new-anchor geometry, with the no-regression KL as the licensed projection.
- (C4) **The convergence-depth law.** EqProp update alignment as a function of settling residual, with routing parameters provably the most sensitive group (their gradient lives in differences of per-operator potentials — plausibly formalizable in your scalar-chain style).
- (C5) **Nonlinear order-independence.** Our `swap_discrepancy` vs. linearized-Gram gap (smoke: median exact discrepancy 0.106 against small linearized interference) suggests the affine obstruction term dominates early; a second-order expansion of the swap formula would make the probe’s bias analyzable.
- (C6) **Itinerant containment.** Clamping recovers scheduled CAROM; symmetrizing  $\rho$  recovers the DEQ — stated in our notes as a proposition, provable in your framework at the linear-phase level.

## 9 Immediate run list

(1) `tau_three_hypothesis_sweep` over the audited run’s saved checkpoints — highest information, zero retraining. (2) `schedule_stability_probe` on the final GPU checkpoint; if the non-uniform-stability smoke reading persists at convergence, decide between Lyapunov-constrained training and certified-schedule execution before any deployment-flavored claims. (3) `cross_slot_coupling_probe` attached to every typed-slot comparison, permanently. (4) Teach-loop reachability check, then KL distillation registration, then re-run paired M2. (5) PC-CAROM staged ignition from the BPTT checkpoints with residual-criterion settling. (6) The belief-register fifth arm, evaluated as portfolio.